Quantification: 548_hyp_HP_z08c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	14
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3376.5	3496	2371.15	px^2
major_axis_length_px	lacuna shape	93.8391	97.983	40.0429	px
minor_axis_length_px	lacuna shape	47.0194	49.0339	20.4227	px
aspect_ratio	lacuna shape	2.1103	1.8291	0.8769	unitless
eccentricity	lacuna shape	0.8243	0.8372	0.1112	unitless
solidity	lacuna shape	0.7964	0.7899	0.0881	unitless
roots_count	headline	12.4286	9.5	7.7332	count
ring_length_r30_px	headline	380.929	373	179.683	px
ring_length_r60_px	reported	683.071	701.5	278.952	px
owned_length_px	ownership-dependent	1074.26	1066.72	639.754	px
edge_count	ownership-dependent	53.2857	55.5	28.8509	count
mean_edge_length_px	ownership-dependent	19.6398	18.6074	3.6393	px
perimeter_px	normalised	303.571	300.856	146.593	px
ring_area_r30_px2	normalised	9778.57	10239.5	3832.1	px^2
ring_area_r60_px2	normalised	20903.1	22486.5	7677.32	px^2
in_frame_fraction_r30	normalised	0.9808	1	0.0436	unitless
in_frame_fraction_r60	normalised	0.9498	1	0.0907	unitless
ring_density_r30	normalised	0.0390364	0.0417665	0.00884613	px^-1
ring_density_r60	normalised	0.0338332	0.0344981	0.00785075	px^-1
roots_per_100px_perimeter	normalised	3.9879	4.3689	1.3928	per 100 px
ring_attached_length_r30_px	network v2	356.857	356.5	180.477	px
ring_attached_length_r60_px	network v2	539.286	588	272.701	px
ring_length_w_r30_px	network v2	435.587	427.212	205.012	px
ring_length_w_r60_px	network v2	781.867	797.068	316.797	px
sholl_crossings_r10	network v2	9.5714	9.5	4.4672	count
sholl_crossings_r20	network v2	9.0714	10	4.3934	count
sholl_crossings_r30	network v2	9	8.5	5.0077	count
loop_count	loops, report only	2.5714	2	2.4088	count
loop_edge_share	loops, report only	0.1333	0.1056	0.1176	share
ring_width_mean_r30_px	width	7.3215	7.3583	0.6259	px

Per field
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measure	value	units
analysed_area_px2	985616	px^2
total_skeleton_length_px	27938	px
canalicular_length_density_per_px	0.0283457	px^-1
junction_count	574	count
junction_density_per_px2	0.00058238	px^-2
lacuna_count	19	count

lacunae_per_px2	1.928e-05	px^-2
skeleton_component_count	357	count
mean_component_length_px	78.2577	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0324614	px^-1
field_density_without_flagged_per_px	0.0283457	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	227	px
unattached_component_count	12	count
canalicular_length_density_attached_per_px	0.0283457	px^-1
field_length_density_with_unattached_per_px	0.028576	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	50	count
loop_edge_share	0.1193	share
edge_count	1492	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6529	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	27938	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1925	65.1309	45.5827	1.4289	0.7143	0.7606	0.0397	28.706	675.968	True
2	3961	122.826	46.3249	2.6514	0.9261	0.8431	0.4753	53.5506	452.253	False
3	547	38.4822	18.6369	2.0648	0.8749	0.9071	1.3303	28.7715	719.596	False
4	4287	134.045	52.6927	2.5439	0.9195	0.6794	-0.8737	66.8097	864.372	False
5	3031	74.7141	59.7338	1.2508	0.6007	0.7786	-0.0729	198.233	105.492	False
6	1249	97.3283	20.9931	4.6362	0.9765	0.6665	0.811	244.751	234.965	False
7	833	42.3386	25.9256	1.6331	0.7906	0.8976	0.935	290.168	253.631	False
8	8164	134.063	85.066	1.576	0.7729	0.8013	-0.8691	359.846	342.936	False
9	5658	108.902	72.88	1.4943	0.7431	0.7761	-0.0527	409.015	685.903	False
10	1968	65.0915	47.7903	1.362	0.6789	0.7977	-1.2589	372.308	23.5991	True
11	2190	104.344	39.1379	2.6661	0.927	0.7344	0.1875	474.681	15.2301	True
12	2786	87.8405	47.2353	1.8596	0.8431	0.8199	0.6054	469.408	431.352	False
13	4506	98.6377	61.8033	1.596	0.7794	0.8877	-0.0114	577.89	721.96	False
14	534	30.0222	23.0801	1.3008	0.6395	0.9287	0.8021	729.066	431.221	False
15	1086	59.0095	32.8062	1.7987	0.8312	0.7126	1.4627	783.261	210.357	False
16	4263	130.21	50.8324	2.5616	0.9207	0.6991	-1.4214	797.485	870.568	False
17	1744	52.1215	44.4583	1.1724	0.522	0.9483	-1.1943	879.793	1002.11	True
18	6366	155.327	60.2608	2.5776	0.9217	0.7519	-1.2656	913.286	921.258	False
19	7862	142.695	80.4282	1.7742	0.826	0.8281	1.2173	982.477	693.681	True

Network per lacuna

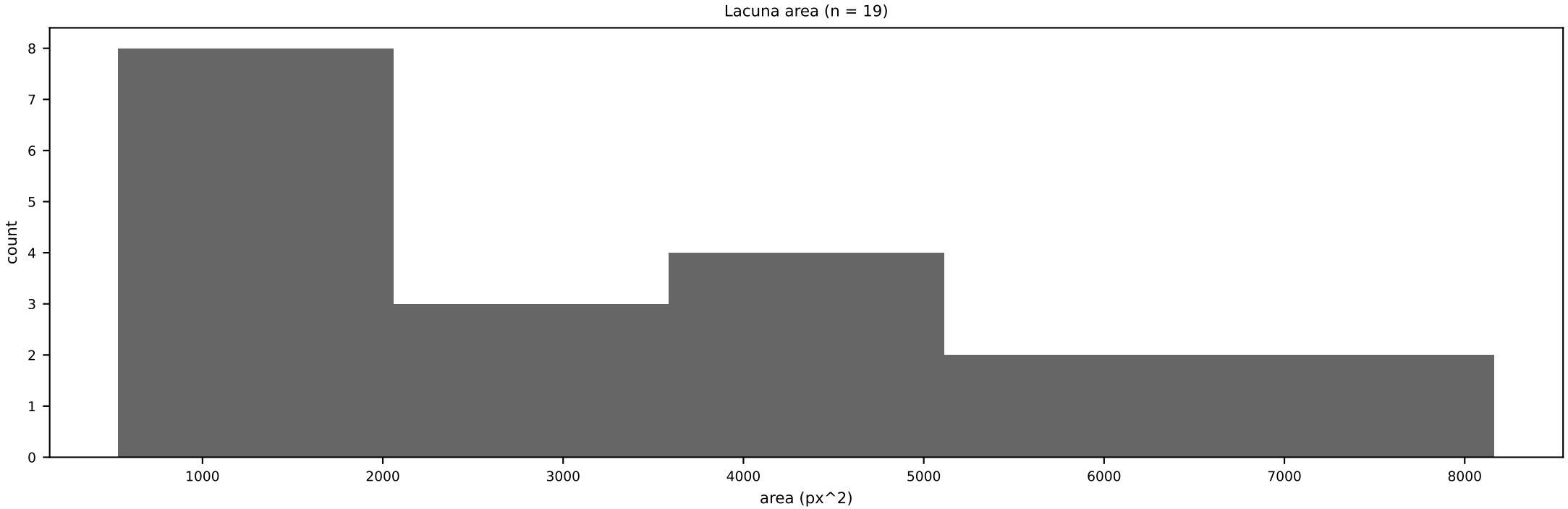
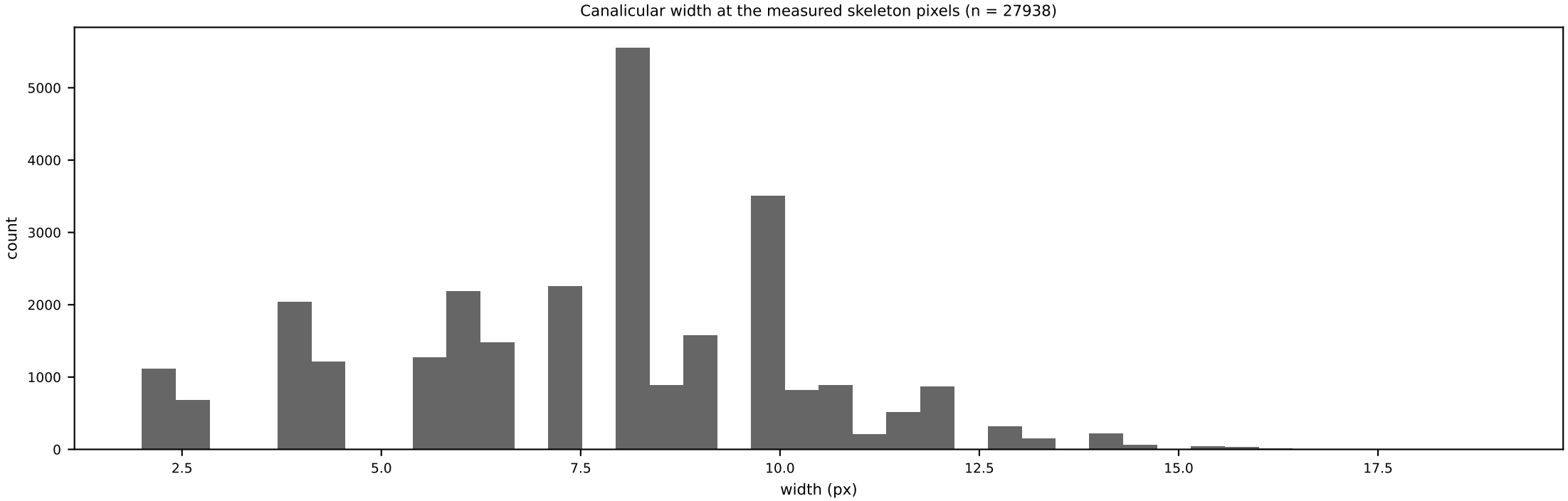
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	207	372	701.311	24	29.2213	6.4041
2	9	242	500	1214.37	67	18.125	8.1995
3	6	174	280	255.723	17	15.0425	6.2003
4	20	567	938	2019.41	94	21.483	7.3796
5	7	437	875	1610.34	85	18.9451	6.95
6	10	329	478	374.635	26	14.409	7.337
7	6	168	273	301.936	17	17.7609	7.7723
8	26	713	1248	1671.94	93	17.9778	7.5994
9	25	505	860	1187.53	65	18.2697	7.0548
10	7	246	392	519.718	34	15.2858	7.2315
11	9	168	358	1437.18	69	20.8287	7.5352
12	6	377	728	2002.65	70	28.6093	8.1913
13	13	369	675	1583.45	73	21.6911	7.5555
14	1	144	445	217.551	10	21.7551	6.6522
15	9	204	535	945.904	40	23.6476	6.7167
16	19	553	923	809.838	46	17.6052	6.7694
17	0	8	50	0	0	0	9.2699
18	17	551	805	844.352	43	19.6361	8.123
19	10	363	661	675.198	43	15.7023	7.1313

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	220.61	5180	11673	0.7389	0.6782	0.0399614	0.0318684	3.173
2	327.706	10056	22309	0.8579	0.7712	0.0240652	0.0224125	2.7464
3	102.326	3157	5234	0.9041	0.7441	0.0551156	0.0534964	5.8636
4	434.274	13343	26739	0.9794	0.8718	0.0424942	0.0350799	4.6054
5	274.007	10299	23608	1	1	0.0424313	0.0370637	2.5547
6	225.38	6965	13598	1	1	0.0472362	0.0351522	4.437
7	121.983	3984	7252	1	1	0.0421687	0.0376448	4.9187
8	522.985	15465	29728	1	1	0.0461041	0.0419806	4.9715
9	451.63	13578	29430	1	1	0.0371925	0.0292219	5.5355
10	224.368	5419	11920	0.697	0.6419	0.0453958	0.0328859	3.1199
11	356.534	5694	11685	0.6133	0.5772	0.0295047	0.0306376	2.5243
12	268.777	10180	22542	1	1	0.0370334	0.0322953	2.2323
13	328.534	11289	25601	1	1	0.0326867	0.0263662	3.957
14	95.6396	5395	16431	1	1	0.0266914	0.027083	1.0456
15	176.066	7548	20526	1	1	0.027027	0.0260645	5.1117
16	441.789	13369	27214	1	1	0.0413644	0.0339164	4.3007
17	173.397	1958	3433	0.6626	0.6178	0.0040858	0.0145645	0
18	478.902	12272	22431	0.9903	0.9107	0.044899	0.0358878	3.5498
19	502.902	11117	23341	0.7175	0.6712	0.0326527	0.0283193	1.9885

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	184	256	237.965	430.889	5	5	4
2	198	273	272.966	576.044	6	5	4
3	164	206	195.439	321.321	3	4	2
4	493	780	644.428	1083.26	12	11	17
5	395	827	498.718	995.021	8	12	14
6	329	411	377.706	540.831	8	11	5
7	136	235	197.823	318.907	4	6	4
8	696	1030	818.625	1421.53	18	19	12
9	499	759	579.357	988.619	13	11	12

10	231	353	277.48	448.576	6	4	6
11	167	308	178.498	396.492	4	5	3
12	359	599	437.304	831.352	14	9	10
13	354	577	417.12	762.784	11	8	13
14	131	194	168.924	518.801	6	3	4
15	162	270	232.309	618.441	5	4	7
16	549	672	630.286	1050.45	13	12	6
17	0	0	8.2426	58.4264	0	0	1
18	531	717	627.215	918.779	13	12	16
19	332	445	420.647	748.097	9	5	9

548_hyp_HP_z08c1-2: distributions, pre-validation, px